

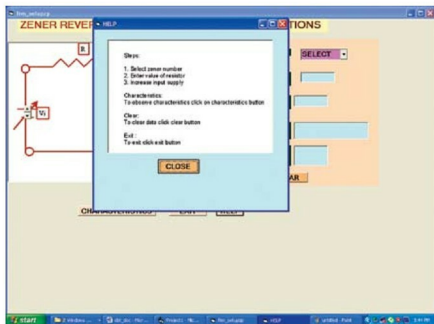


Fig. 3: Help menu screen

zener are calculated using the calculation() function.

The Cmd_clear_Click() function is used to clear all the input parameters.

When you click the executable file of this program, you will see a window with a circuit diagram for drawing the characteristics of a zener diode (refer Fig. 1). Select the zener diode from the list given in the 'select' combo box. Enter the value of the resistor in the respective box and then increase the input voltage using a vertical scroll-bar control as explained in the testing procedure given below. As you increase the input voltage, the corresponding current and voltage across the zener diode are displayed. When the current reaches the maximum value, a warning message is displayed not to increase the voltage further.

Another warning message is displayed if you try to increase the input voltage without selecting the zener diode or without entering the value of the resistor. At any point, if you want to change the zener diode, clear all the current parameters using 'clear' button. Then you can perform the next experiment by selecting another zener diode.

A 'characteristics' button is provided to observe the characteristics of the selected zener diode. After clicking it, a 'characteristics' window opens and the reverse characteristic of the zener diode is displayed as shown in Fig. 2.

To know the coordinates of any point on the characteristics graph, click the left mouse button on the graph. To close the characteristics window, use 'close' button.

A 'help' window is displayed when you click 'help' button. The steps to perform the experiment are shown in Fig. 3. Close 'help' window using 'close' button and exit the program with 'exit' button.

Design steps

1. Launch Visual Basic from the desktop
2. Select 'standard exe' option from 'new project' window. You will see Form1 in the next screen
3. Place the tools on Form1 as shown in the program output (Fig. 1)
4. Write the program by defining the various object controls in the form
5. Click 'start' button under 'run' menu to compile the code
6. Save the program by giving it a suitable filename. To create an

executable (.exe) file of the program, go to 'file' menu and click 'make .exe' option

Testing procedure

1. Click the executable file. The program output screen appears as shown in Fig. 1

2. Increase the input voltage (V_i) by clicking the vertical scroll bar on the left-hand side of the circuit diagram. It will prompt you to enter the value of resistor R on the right-hand side of the screen

3. Enter the resistor value, say, 200, and increase the input voltage. The screen should display the "select zener" message. Click 'ok' button

4. Select zener 1N4733A by clicking the combo box. You can select any zener diode for your experiment by referring to the zener diode ratings table

5. Start increasing the input voltage; observe the input voltage, current value and voltage across the zener in corresponding edit boxes. As V_i increases from 0V, voltage across zener (V_z) also increases. But current I_z is negligible and constant at 0.01 mA

When V_i is increased beyond 5.1V, V_z becomes constant at 5.1V but I_z suddenly increases to 0.5 mA and keeps on increasing. When V_i is 6.5V, I_z is 7 mA. You can verify current I_z practically by wiring the circuit on a breadboard and measuring the current using an ammeter at the point 'I' in the circuit. The practical value of I_z is found to be 6.7 mA, which is quite acceptable

6. When the input voltage reaches 15V, the message "Stop—do not increase input" is displayed. Click 'ok' button. The textbox values for the input voltage, current and voltage across the zener diode should change to zero

7. Click 'clear' button and all textboxes must clear

8. Clicking 'characteristics' button should open 'reverse characteristics' window. Click the left mouse button on the graph area. The correspond-